Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Initial position

Meaning in my words How would you explain initial position to a friend?	**How can you tell?** At 0 s the cart is beside the +5 m mark. Where does its position graph begin?
Example or sketch The cart starts at +5 m and moves 2 m left. What two positions would you mark?	**Non-example and why** “The clock starts at zero, so the cart starts at zero.” What do the two zeros mean?

Bound

Meaning in my words How would you explain bound to a friend?	**How can you tell?** For 10 ± 2 km, what is the farthest allowed position on either side?
Example or sketch Mark the bounds for a forecast of 15 ± 3 km.	**Non-example and why** A student labels 2 km as the upper endpoint of 10 ± 2 km. What does the 2 mean?

Frayer models

SECTION A / VOCABULARY Name _____

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Half-width

Meaning in my words How would you explain half-width to a friend?	**How can you tell?** A model allows 8 to 12 km. How far is either endpoint from 10 km?
Example or sketch How would you write a range centered on 20 km with half-width 3 km?	**Non-example and why** Someone calls 6 km the half-width of 17 to 23 km. What did they measure?

Assumption

Meaning in my words How would you explain assumption to a friend?	**How can you tell?** What must be true to predict another 2 m each second from a cart's earlier motion?
Example or sketch Write an assumption beside a straight-line forecast for XJ.	**Non-example and why** Is tomorrow's camera reading an assumption once we measure it?

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Draw a center tick and both endpoints. Show how starting-position and velocity bounds contribute to the final half-width.

Choose zero, direction, and scale.



My marks mean

On assignment: Bound

SECTION E / BACK / FINAL MEETING ONLY Name _____

Two XJ teams predict a center of 20 km. Team A gives ±1 km; team B gives ±4 km. A telescope has one narrow field.

PICK A JOB / DELIVER ONE PRODUCT

☐ **Survey engineer**

Draw both possible bands on one axis. Recommend which report requires a wider search; give both sets of endpoints.

☐ **Weather-desk presenter**

Produce a forecast graphic with two bands. Explain to a viewer which search is wider and what the graphic cannot tell us about impact.

One partner makes the graphic; one delivers the first call. Show your evidence to the crew or the audience.

OUR GRAPHIC / COMMAND OR ON-AIR CUE

INCOMING UPDATE / WAIT FOR YOUR TEACHER

Listen to the new report. Switch speakers. Keep or change your call; point to the new evidence.

Editor's check: Distinguish an input bound from a change in the assumed motion.